



FACUNDO JOEL ALLIA FERNANDEZ

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Portfolio: facundoallia.github.io

Data Scientist and Python Developer with 6+ years bridging advanced analytics, machine learning, and financial engineering. Specialized in building production-grade data solutions—from ETL pipelines and automated systems to pricing models and risk analytics. Expert in Python, SQL, and cloud platforms with proven track record transforming complex financial data into actionable insights across Big Four firms, investment banking, and derivatives markets. Combines deep quantitative finance expertise with modern ML/AI implementation and scalable data architecture.

WORK EXPERIENCE

Banco Santander Corporate & Investment Banking - Data Scientist & Automation Engineer | 2025

- Design and implement Python-based automation solutions replacing manual workflows, reducing operational costs and error rates. Develop ETL pipelines ingesting data from multiple sources including data lakes and Murex systems.
- Built comprehensive Python solution analyzing bank exposure across multiple benchmarks with custom GUI for functional analysts, streamlining decision-making processes.
- Engineered full-stack application automating structured notes management, replacing manual Excel processes for entire front-office team—significantly improving efficiency and data accuracy.
- Develop Power BI dashboards for monitoring automated processes and presenting analytical results to stakeholders.

Technologies: Python, SQL, Databricks, Murex, Power BI, Power Automate, Azure, ETL/ELT pipelines, RAG/LLMs

PricewaterhouseCoopers (PwC) - Quantitative Analyst | 2024 – 2025

- Specialized in complex derivatives pricing using stochastic models and Monte Carlo simulation. Developed credit derivatives valuation engine with Python implementing CVA/DVA frameworks.
- market risk assessments and built XVA calculation models for regulatory compliance, collaborating with audit teams on validation frameworks.
- Regular contributor to technical newsletter analyzing Decentralized Finance trends and their impact on traditional markets, leveraging blockchain analytics and quantitative research.
- Performed scenario analysis and stress testing for financial instruments using Python, R, and proprietary risk systems.

Technologies: Python, R, Excel, Power BI, Azure, Murex, Bloomberg Terminal, Reuters Terminal

Accenture - Financial Data Transition Specialist | 2022 – 2023

- Developed KPIs and financial metrics for enterprise reporting, identifying process optimization opportunities boosting profitability.
- Implemented forecasting tools and management software improving financial visibility and decision-making.
- Worked with SAP HANA and cloud platforms for large-scale data integration and analytics.

Technologies: SAP, SAP HANA, AWS, Azure, Microsoft Excel, SharePoint

Nixtla - Data Scientist | 2022

- Performed time series forecasting and modeling (ARIMA, ETS, CES) using Python and R for predictive analytics.
- Developed automated forecasting pipelines on Databricks with PySpark for scalable model deployment.
- Authored technical documentation and conducted model performance evaluations ensuring optimal accuracy.

Technologies: Python, R, SQL, PySpark, Databricks, AWS, Docker, APIs, Git/GitHub

Matba Rofex - Institutional Derivatives Trader - Market Making | 2021 – 2023

- Actively traded futures and options on financial assets and commodities, managing market-funded account to meet volume standards.
- Demonstrated strong quantitative skills forecasting market trends, **consistently outperforming benchmark for two consecutive years.**
- Developed Python-based trading analytics and backtesting frameworks for strategy optimization.

Technologies: Python, SQL, APIs, Trading platforms (Matriz, Etrader), TradingView

Logos Servicios Financieros - Financial Advisor & Founder | 2020 – Present

- Provide financial advisory services and portfolio management with customized investment strategies.
- Established and managed proprietary client portfolio ensuring high-quality service and satisfaction.

Libertad S.A. - Commercial Data Analyst | 2021 – 2022

- Developed monthly/annual budgets and demand forecasts incorporating macroeconomic variables via interactive dashboards.
- Implemented real-time sales data ingestion and analysis solution at scale, adding significant operational value.

Technologies: Python, SQL, SAP, Power BI, Microsoft Excel, proprietary ERP/CRM

Peugeot Avant Autocity - Data Analyst | 2019

- Maintained and structured operational databases using SQL, Excel, SAP and proprietary ERP/CRM systems.
- Built dashboards and performance reports supporting business decisions.

Technologies: SQL, Microsoft Exel, SAP, Power BI, proprietary ERP and CRM.

STUDIES AND CERTIFICATIONS

- **Master's in Finance** - EAE Business School Madrid (2024) - *Minor in Smart Money & Start Ups*
- **Postgraduate Degree in Quantitative Finance** (Quant Program) - UCEMA (2022)
- **Data Science Coding Bootcamp** (+700h.) - Henry (2022)
- **Financial Markets Specialization** (Derivatives trading focus) - Bolsa de Comercio de Rosario (2021)
- **Bachelor of Business Administration** (Finance Specialization) - Universidad Nacional de Córdoba (2020)
Teaching assignment in Stock Exchanges and Stock Market
- Financial Market Suitability and Stock Market Producer Registration N° 1271 - CNV (2021)

Courses and certifications:

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|---|---------------------------------------|
| - Cloud Practitioner Essentials - AWS | - Time Series Forecasting - Kaggle |
| - Strategic thinking - CAME | - Digital Disruption in Finance – UNC |
| - Personal brand - Platzi | - Regulatory Consulting – PWC |
| - Harvard Negotiation Method - IEP | - PLAFT- Balanz University |
| - Digital Marketing Strategies - CAME | - Personal finance - UNAM |
| - Python for Finance, Investment & Data Analytics – Udemy | - Advanced Excel - IPAP |

TECHNICAL SKILLS

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| - Python: Numpy, Pandas, SciPy, and visualization libraries. | - SQL |
| - Machine Learning: TensorFlow and Scikit-Learn. | - Amazon Web Services |
| - PySpark | - Power BI |
| - Git - GitHub | - APIs |
| - Blockchain, cryptocurrencies, and DEFI experience | - Docker |
| - Expert in Microsoft Office | - Financial modeling (TSF, risk, etc.) |
| - Knowledge of probability and statistical methods | - R language |
| - Financial Valuation | - LLMs, agents, RAGs and AI tools |
| - Bloomberg Terminal | - Reuters Terminal |
| - Murex | - Power automate/apps |

LANGUAGES

- **Advanced English**
 - EF SET English Certificate 77/100 (C2 PROFICIENT) - EF Standard English Test
 - Business English Certificate Vantage (CEFR LEVEL B2) - Cambridge University
- **Basic Portuguese** (A2) - Business oriented
- **Native Spanish**

SOFT SKILLS

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| - Critical analysis and problem-solving skills | - Collaborative and cooperative teamwork |
| - Effective communication and negotiation skills | - Meticulous attention to detail |
| - Ability to work effectively under pressure | - Adaptability to changing situations |
| - Self-learning and continuous skill development | - Results and goal-oriented mindset |

PUBLICATIONS

- Allia Fernandez, F. J. “Construction of risk-neutral probability density functions through butterfly spreads”. BCR. Jun 30, 2022. [Available at SSRN](#)
- Allia Fernandez, F. J. “Stock classification using k-means clustering”. UCEMA. Dec 7, 2022. [Available at SSRN](#)